

UPSC PRELIMS 2026: ESSENTIAL ECOLOGY CONCEPTS

ECOSYSTEM COMPONENTS

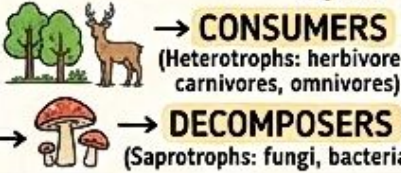
ABIOTIC (Non-living):
Climate, Temp, Water,
Soil, Nutrients



BIOTIC (Living):

PRODUCERS

(Autotrophs: plants, algae)



CONSUMERS

(Heterotrophs: herbivores,
carnivores, omnivores)

DECOMPOSERS

(Saprotrophs: fungi, bacteria)

FOOD CHAIN & FOOD WEB

FOOD CHAIN

Linear energy flow

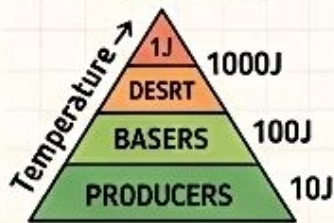


Grass → Grasshopper → Frog

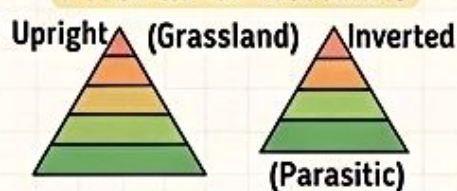
ECOLOGICAL PYRAMIDS

PYRAMID OF ENERGY

Always UPWRIGHT
(base: producers,
top: predators,
1000J → 100J →
→ 10J → 1J).



PYRAMID OF NUMBERS



PYRAMID OF BIOMASS



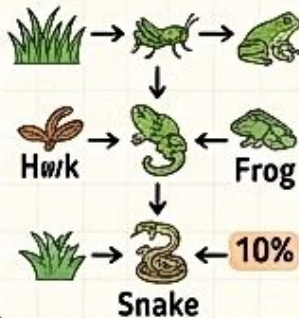
ABHISHA AI

by Abhishek Pathak

FOOD CHAIN

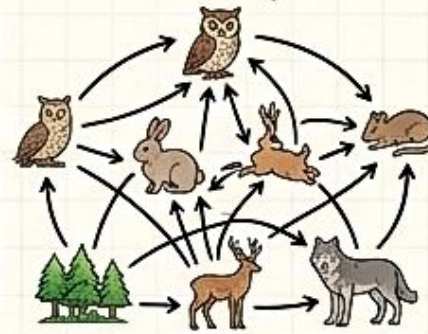
Linear energy flow

10%



FOOD WEB

Interconnected chains,
stability



BIODIVERSITY LEVELS

GENETIC DIVERSITY

Variation within
species



SPECIES DIVERSITY

Number & Evenness



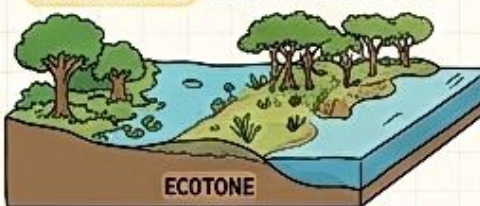
ECOSYSTEM DIVERSITY

Habitat variety



ECOTONE & EDGE EFFECT

ECOTONE: Transition zone



ECOTONE

(e.g., Mangrove between
between Land & Sea, Marsh).
High biodiversity.

EDGE EFFECT

Increased species
richness at boundary.

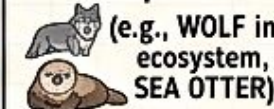
BIOMES OF THE WORLD



KEYSTONE & FLAGSHIP SPECIES

KEYSTONE

Disproportionate
impact



FLAGSHIP

Iconic for
conservation



INDICATOR

Air quality



ECOLOGICAL SUCCESSION

PRIMARY SUCCESSION

Bare rock → Lichens → Mosses → Grasses → Shrubs
→ Trees → Climax Community (long time).

SECONDARY SUCCESSION

Disturbed area (e.g., Fire) → Grasses → Shrubs → Trees
→ Climax Community (faster).

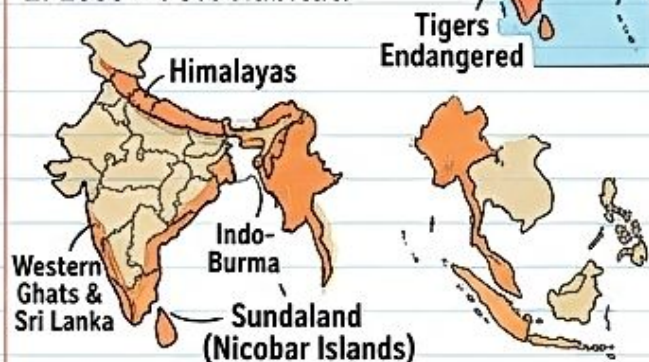
INDIAN WILDLIFE CONSERVATION: UPSC PRELIMS 2026

ABHISHA AI by Abhishek Pathak

1. BIODIVERSITY HOTSPOTS (CONSERVATION INTERNATIONAL)

CRITERIA:

- 1,500+ Endemic Vascular Plants (>0.5% World Total).
- Lost $\geq 70\%$ Habitat.



6. BIONSERVATION STRATEGIES

- 18 BIOSPHERE**
- Connect in natural habitat.
 - 12 under UNESCO MAB (Ex: Nilgiri, Sunderbans, Panna)
- EX-SITTU ZONE**
- Flatact reoorion, Roseranle Tourism, Bihaniel, Grasha)

2. IUCN RED LIST CATEGORIES

Critically Endangered (CR):

90%+ Decline (10 yrs) or <50 Individuals



Tiger (EN)

Endangered (EN):

70%+ Decline or <250



Gharial (CR)

Vulnerable (VU):

50%+ Decline or <1000

*IUCN GREEN STATUS assesses recovery

5. CONSERVATION STRATEGIES

IN-SITU (ON-SITE):

Protected in natural habitat.

(Ex: NP, WLS, BR)

EX-SITU (OFF-SITE):

Outside habitat.

(Ex: Zoo, Botanical Garden, Gene Bank)

PROJECTS:

- **PROJECT TIGER** (54th Reserve: Rani Durgavati)
- **PROJECT ELEPHANT** (Merged with PTE Division)
- **PROJECT CHEETAH** (Kuno NP, Expansion to Gandhi Sagar & Shahabad, F1 Generation born)
- **OTHER: PROJECT LION** (Gir & Barda)
- **PROJECT DOLPHIN**
- **PROJECT GIB**

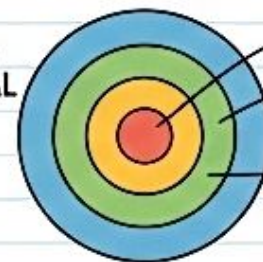
3. PROTECTED AREA NETWORK (WILDLIFE PROTECTION ACT, 1972)

NATIONAL PARKS (NP)	High Protection No human activities/grazing Ex: Jim Corbett, Kaziranga
WILDLIFE SANCTUARIES (WLS)	Limited rights for local people Ex: Gir, Periyar

TOTAL NP in INDIA: ~107 (as of 2026)

4. BIOSPHERE RESERVES & ZONATION

18 TOTAL



CORE ZONE (Strictly Protected)

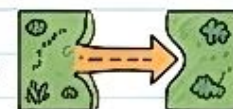
BUFFER ZONE (Research, Education, Sustainable Tourism)

TRANSITION ZONE (Settlements, Agriculture, Management)
(Ex: Nilgiri, Sunderbans, Panna)

6. WILDLIFE CORRIDORS & ESZ

WILDLIFE CORRIDORS:

- Connect fragmented habitats.
- Prevent conflicts.
- Over 150 Elephant Corridors (highest in West Bengal, Odisha)



HABITAT PATCHES



HABITAT PATCHES

ECO-SENSITIVE ZONES (ESZ):

- Shock absorbers.
- Transition from PA to non-PA.
- Up to 10 km around NP/WLS.

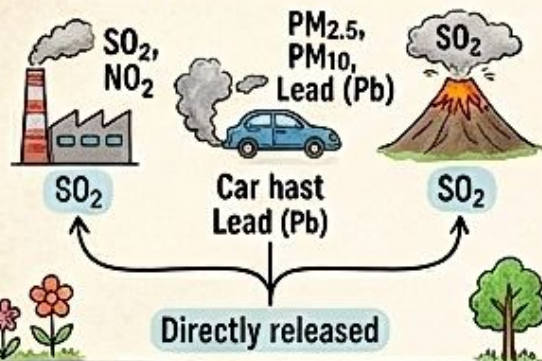
ESSENTIAL ENVIRONMENTAL POLLUTION CONCEPTS FOR UPSC PRELIMS 2026

Curated by Abhishek Pathak, ABHISHA AI

SECTION 1: AIR POLLUTANTS: PRIMARY VS SECONDARY

PRIMARY POLLUTANTS:

Direct Emissions from Source



SECONDARY POLLUTANTS:

Atmospheric Reactions

Primary Pollutants + Sunlight + H₂O

GROUND-LEVEL OZONE (O₃)

SMOG

ACID RAIN (H₂SO₄, HNO₃)

PEROXYACETYL NITRATE (PAN)



SECTION 2: WATER POLLUTION SOURCES

WATER POLLUTION: POINT & NON-POINT SOURCES

POINT SOURCE:

Identifiable, Single Location



Industrial Effluents



Sewage Treatment Plants

Major Pollutants: Sewage, Metals, Chemicals, Pathogens, Thermal Pollution

NON-POINT SOURCE:

Diffused, Wide Area



Agricultural Runoff (Fertilizers/Pesticides)

Urban Stormwater

Soil Erosion



AGRICULTURAL RUNOFF, SEWAGE, and SEDIMENT

SECTION 3: EUTROPHICATION CYCLE

1 Nutrient Enrichment:

Nitrates (N) & Phosphates (P) from Agriculture/Sewage enter water

2 Algal Bloom:

Rapid algae growth on surface blocks light.

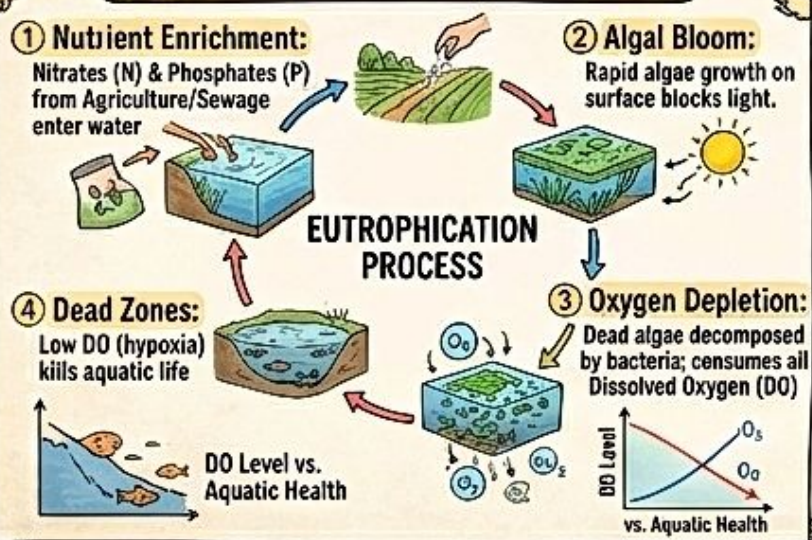
3 Oxygen Depletion:

Dead algae decomposed by bacteria; consumes all Dissolved Oxygen (DO)

4 Dead Zones:

Low DO (hypoxia) kills aquatic life

EUTROPHICATION PROCESS



SECTION 4: NOISE POLLUTION STANDARDS (INDIA)

NOISE POLLUTION (REGULATION & CONTROL) RULES, 2000



SOLID WASTE MANAGEMENT (SWM) RULES



Rules monitor and set limits for different zones

Promotes 3Rs: Reduce, Reuse, Recycle

SECTION 6: PLASTIC POLLUTION MANAGEMENT

PLASTIC WASTE MANAGEMENT: RULES & STRATEGIES

SINGLE-USE PLASTIC (SUP) BAN

Prohibited: Straws, cutlery, thin bags (<120 microns)

- Prohibited: Straws, cutlery, thin bags (<120 microns)
- Prohibited: Ltems, Cliters, Hazardouss, Cornons
- Prohibited: Bags, Plastloss

EXTENDED PRODUCER RESPONSIBILITY (EPR)

Producer/Brand Owner liable for collect-back & recycling of plastic packaging



PWM Rules 2024: Stricter labeling for biodegradable/compostable plastics to prevent 'greenwashing'

UPSC PRELIMS 2026: CLIMATE CHANGE FUNDAMENTALS & MECHANISMS

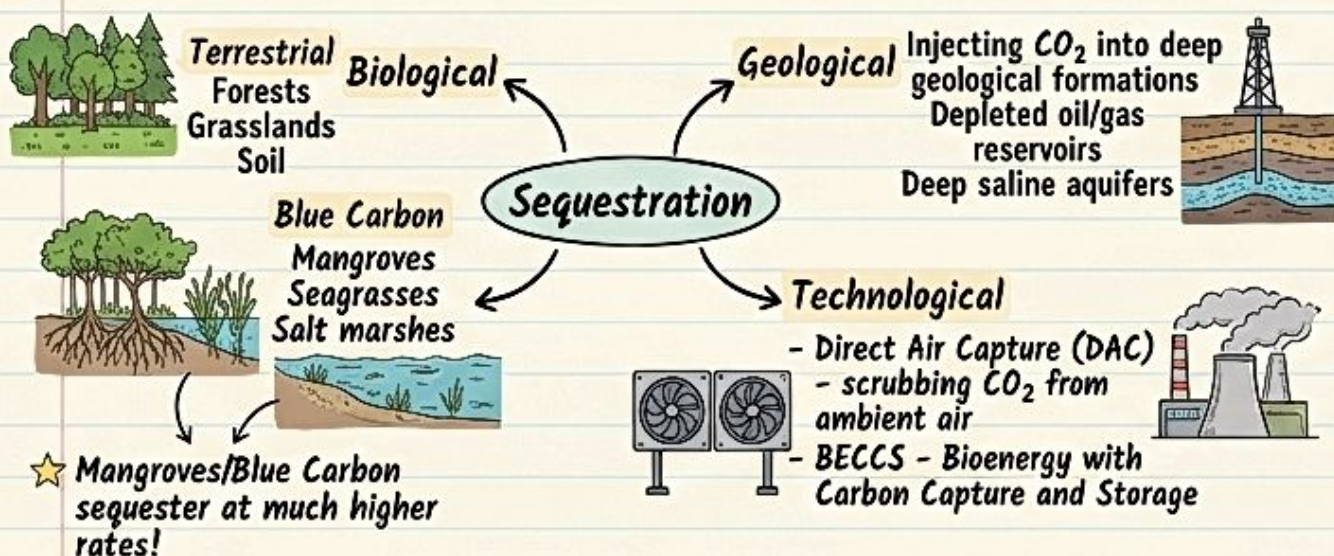
1 GREENHOUSE GASES & GWP

Gas	Major Sources	100-yr GWP	Atmospheric Lifetime
Carbon Dioxide (CO_2)	Fossil fuels, deforestation	1 (Ref)	100s-1000s yrs
Methane (CH_4)	Rice paddies, livestock, landfills	~28-30	~12 yrs.
Nitrous Oxide (N_2O)	Nitrogen fertilizers, industry	~273	~100+ yrs.
HFCs	Refrigeration, ACs	1,000s	Varies
Sulfur Hexafluoride (SF_6)	Electrical switchgear	~24,000+	3,200 yrs.

Key Concepts

★ **UPSC TIP:** Kigali Amendment (2016) targets HFC phase-down, even though they are non-ozone depleting. India tightening norms for production.*

2 CARBON SEQUESTRATION: REMOVING CARBON FROM ATMOSPHERE



3 CARBON TRADING & MARKETS: KYOTO TO PARIS

A. KYOTO PROTOCOL (OLD REGIME)



3 Flexible Mechanisms:

- Clean Development Mechanism (CDM):** Developed countries fund projects in developing countries (e.g., India) to get credits
- Joint Implementation (JI):** Projects between two developed countries.
- International Emissions Trading:** Trading Assigned Amount Units (AAUs)

B. PARIS AGREEMENT (CURRENT REGIME - ARTICLE 6)

• **Article 6.2:** Bilateral/Voluntary Internationally Transferred Mitigation Outcomes (ITMOs).

*Ex: India-Japan JCM (Joint Crediting Mechanism).

• **Article 6.4:** Creates new centralized UN mechanism to replace CDM

Crucial

'CORRESPONDING ADJUSTMENT' RULE:
To prevent double counting, selling credit means the seller cannot count it towards its own NDC.

4 INDIA-SPECIFIC UPDATES (FOR PRELIMS)

- **Carbon Credit Trading Scheme (CCTS):** India launched domestic compliance market. Governed by Bureau of Energy Efficiency (BEE).
- **Target Sectors:** Heavy industries like cement, aluminum, petroleum refineries, textiles. Power sector excluded in first phase (FY 2025-26).
- **Prakriti 2026:** Major int'l conference in New Delhi, focus on India as high-quality credit hub.
- **Green Credit Programme (GCP):** Distinct from carbon credits; incentivizes water conservation, afforestation, etc. beyond carbon.

GLOBAL CLIMATE CHANGE FRAMEWORKS & CONVENTIONS

1 UNFCCC (The Mother Convention) 1992



Establishment (Rio Earth Summit, 1994 Entry)



Objective: Stabilize GHG concentrations,
Prevent dangerous human interference



Key Principle: CBDR-RC (Common But Differentiated Responsibilities)



Country Classification

(Annex I: Developed; Annex II: Finance; Non-Annex I: Developing)

2 KYOTO PROTOCOL vs. PARIS AGREEMENT

(Key Shift)

FEATURE	KYOTO (1997)	PARIS (2015)
Approach	Top-Down / Bottom-Up	Bottom-Up
Target Group	Annex I	All Countries
Legal Status	Binding targets	Hybrid: Binding process NDCs not binding
Goal	Reduce specific GHGs	Well below 2°C, aim 1.5°C
Current Status	Ended 2020	Primary framework now

3 REDD & REDD+



REDUCING EMISSIONS FROM DEFORESTATION & FOREST DEGRADATION

→ REDD+ →

1. Conservation of carbon stocks
2. Sustainable management of forests
3. Enhancement of carbon stocks

- Mechanism: Results-Based Payments
- Distinction: Results-Based Payments

REDD+ (UNFCCC)
vs.
UN-REDD Programme
(FAO, UNDP, UNEP)

4 RECENT COP MEETINGS (Key Updates for 2026)

COP 28 (DUBAI, 2023)

1. GLOBAL STOCKTAKE (GST):
First progress assessment.
2. FOSSIL FUEL TRANSITION:
Call to 'transition away'.
3. LOSS & DAMAGE FUND:
Operationalized.

COP 29 (BAKU, 2024)

- The 'FINANCE COP'
- Focused on NEW
COLLECTIVE
QUANTIFIED GOAL
(NCGQ) replacing
\$100B pledge.

COP 30 (BELÉM, 2025)

- Adoption of BELÉM
PACKAGE
- India welcomed JUST
TRANSITION MECHANISM
- TFIF: \$125B Tropical
Forest Investment Fund.



KEY TERMS FOR PRELIMS



NDCs (Updated every 5 years, India NET ZERO by 2070)
GGA (Global Goal on Adaptation)
CARBON CREDIT MARKETS (Paris Art. 6 Rules)



'ENVIRONMENTAL LEGISLATION IN INDIA: KEY ACTS FOR UPSC PRELIMS 2026'

ENVIRONMENT PROTECTION ACT (EPA), 1986

Umbrella Legislation
(Post-Bhopal Gas Tragedy, Stockholm Conf.)

KEY FEATURES:

- (1) Umbrella Framework for Water/Air Acts.
- (2) Central Govt Power (set standards, inspections, manage hazardous substances).
- (3) **EIA NOTIFICATION** Legal Validity from EPA.
- (4) **ECO-SENSITIVE ZONES (ESZ)** Declaration around PAs.



WILDLIFE PROTECTION ACT (WPA), 1972

Protects wild animals, birds, and plants

KEY FEATURES:

- (1) Protected Areas (NPs, WLS, Conservation & Community Reserves).
- (2) **RECENT AMENDMENT:** Rationalized Schedules to 4. Schedule I (Highest protection), Sch II (Protected), Sch III (Plants), Sch IV (CITES species).
- (3) Invasive Alien Species Regulation.
- (4) **VERMIN:** Notified by Centre (No specific Sch).
- (5) Controversial: Religious use transfer of Elephants.

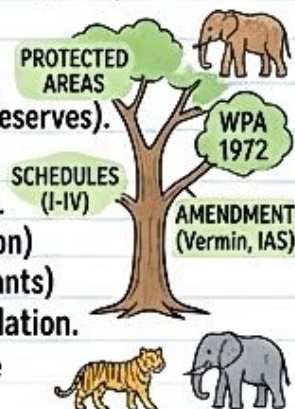


WILDLIFE PROTECTION ACT (WPA), 1972

Protects wild animals, birds, and plants

KEY FEATURES:

- (1) Protected Areas (NPs, WLS, Conservation & Community Reserves).
- (2) **RECENT AMENDMENT:** Rationalized Schedules to 4. Schedule I (Highest protection), Sch II (Protected), Sch III (Plants), Sch IV (CITES species).
- (3) Invasive Alien Species Regulation.
- (4) **VERMIN:** Notified by Centre (No specific Sch).



FOREST CONSERVATION ACT (FCA), 1980

Restricts diversion of forest land for non-forest purposes

KEY FEATURES:

- (1) Applies **ONLY** to notified forests or recorded after 1980.
- (2) **RECENT AMENDMENT** (Van Adhiniyam):
EXEMPTIONS:
 - 100km of borders (strategic linear projects);
 - Security infrastructure (up to 10ha; 5ha)
 - LWE areas (5ha).
- (3) **NON-FOREST PURPOSE** excluded: Zoos, Safaris, Eco-tourism.



EXEMPTIONS

NEW EXCLUSIONS

7 Laws (e.g., EPA, Water/Air Acts) Wildlife Protection Act and IFA, 1927

NATIONAL GREEN TRIBUNAL (NGT) ACT, 2010

Specialized Judicial Body for Environment.
(Article 21 basis)



KEY FEATURES:

- (1) **COMPOSITION:** Chairperson (Retd SC Judge/HC CJ), Judicial & Expert Members.
- (2) **JURISDICTION:** 7 Laws (e.g., EPA, FCA, Water/Air Acts) **EXCEPT** Wildlife Protection Act (WPA), 1972 and IFA, 1927.
- (3) Natural Justice, not CPC. Pollution Pays & Sustainable Development principles.



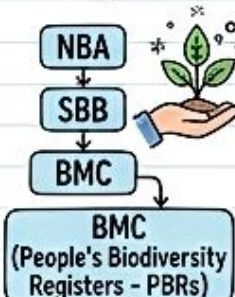
BIOLOGICAL DIVERSITY ACT, 2002



Implements Convention on Biological Diversity (CBD)

KEY FEATURES:

- (1) **THREE-TIER STRUCTURE:** NBA (National - Chennai), SBB (State), BMC (Local).
- (2) **ACCESS & BENEFIT SHARING (ABS):** From genetic resources & traditional knowledge.
- (3) **AMENDMENT:** Exemptions for AYUSH practitioners from ABS fee.



CRITICAL ENVIRONMENTAL LAWS OF INDIA (FOR UPSC PRELIMS 2026)

DETAILED OVERVIEW & KEY AMENDMENTS

ENVIRONMENT PROTECTION ACT (EPA), 1986

UMBRELLA LEGISLATION

OVERVIEW & KEY PROVISIONS

- Enacted under ARTICLE 253 (implementing Stockholm Conference, 1972).
- Empowered Central Govt to establish authorities (e.g., Central Pollution Control Board).
- Formulates standards for emissions/effluents.
- Statutorily backs Environmental Impact Assessment (EIA).
- Declares ECO-SENSITIVE ZONES (ESZ) (Note: Not under Wildlife Act)



FOREST CONSERVATION ACT (FCA), 1980 (AMENDED 2023)

MAJOR 2023 AMENDMENTS

- Applicability: Only to land notified/recorded as forest AFTER 1980
- STRATEGIC EXEMPTIONS: Forest land within 100 km of international borders for national security projects (e.g., roads) are exempt from central clearance.
- EXCLUDED ACTIVITIES: Zoos, safaris, eco-tourism are 'NON-FOREST PURPOSES' - simplified process, fewer clearances required
- Penalties enhanced. → New nec concepts



WILDLIFE PROTECTION ACT (WPA), 1972 (AMENDED 2022)

RATIONALIZED SCHEDULES & KEY AMENDMENTS

OLD SCHEDULES (I-VI)



Vermine (V), Plants (VI) existed

NEW SCHEDULES (I-IV)

SCHEDULE I: HIGHEST PROTECTION (Endangered Sp.) (Tigers, Rhinos)

SCHEDULE II: HIGHER PROTECTION (Commoner mammals, birds)

SCHEDULE III: PROTECTED PLANT SPECIES

SCHEDULE IV: CITES SPECIES (New addition)

- National Board for Wildlife (PM chaired)
- NTCA (National Tiger Conservation Authority)
- Wildlife Crime Control Bureau (WCCB)
- 2022 amendment simplifies Vermin declaration



COMPARISON OF PENALTIES & POWER

ACT	PRIMARY AUTHORITY	NATURE OF OFFENSE & POWER
EPA 1986	Central Govt	Criminal (incl. Jail)
WPA 1972	NTCA, WCCB, Court	Criminal (incl. long Jail)
NGT 2010	Judicial/Expert Members	Civil (Compensation, Binding Orders)
Biodiversity 2023	NBA / SBB	Civil (Heavy Fines, no Jail)

NATIONAL GREEN TRIBUNAL (NGT) ACT, 2010

POWERS, JURISDICTION & MANDATE

- SPECIALIZED BODY for expeditious disposal of env. cases.
- MANDATE: DISPOSE OF CASES WITHIN 6 MONTHS.
- JURISDICTION over 7 ACTS (Water, Air, EPA, Forest, Biodiversity, etc.)
- NO JURISDICTION OVER:
 - Wildlife Protection Act,
 - Forest Rights Act.
- POWER: Grant relief, compensation, restitution. Orders binding.



BIOLOGICAL DIVERSITY ACT, 2002 (AMENDED 2023)

3-TIER STRUCTURE & KEY AMENDMENTS



- 2023 AMENDMENT HIGHLIGHTS:
 - Exemption: AYUSH Practitioners exempt from prior intimation to SBBs for biological resources for traditional medicine.
 - DECriminalIZATION: Violations are now CIVIL OFFENSES (fines) not criminal (jail).
 - FDI rules simplified for research involving Indian bio-resources.

